

Suppression of Current Quantization Effects for Precise Current Control of SPMSM Using Dithering Techniques and Kalman Filter

Hongzhong Zhu, *Student Member, IEEE*, and Hiroshi Fujimoto, *Member, IEEE*

Abstract—High-precision current control is demanded for many mechatronic systems to improve the static and dynamic performances. However, current measurement error inherent in the measurement system degrade the control accuracy, since the motor currents are not guaranteed to follow the desired references. In this paper, quantization error caused by analog-to-digital converters is studied. The dithering techniques combined with Kalman filter are proposed to suppress the quantization effects. First, two dithered systems, including the subtractively dithered and nonsubtractively dithered systems, are designed to whiten the quantization error. In the design of nonsubtractively dithered system, the probability characteristic of the metering noise is utilized to minimize the measurement noise level. Then, Kalman filter is designed to estimate the real current signals from the dithered current measurements. Since the variance of the total measurement error is theoretically calculable according to the proposed dithering techniques, Kalman gain can be determined analytically. Finally, simulations and experiments are performed to verify the effectiveness of the proposed approaches using a high-precision positioning system.

Index Terms—Analog-to-digital converter (ADC), current control, dithering techniques, Kalman filter, motion control, quantization.

I. INTRODUCTION

PRECISE current-controlled permanent magnet synchronous motor (PMSM) drive systems require accurate current measurement to produce high static and dynamic performances [1]–[3]. However, current measurement errors caused by current sensors and analog-to-digital converters (ADCs) limit the current control performance, since the real current signals are not guaranteed to follow the desired references accurately. This would induce torque ripples and deteriorate the control performance especially when a drive system works at low load [4], [5]. Furthermore, current quantization error may also affect the accuracy of rotor position estimation when high-frequency signal injection method is applied for motor sensorless control [6], [7]. Therefore, understanding and suppressing the effects caused by the current measurement errors have attracted a great deal of attention [8], [9].

Fig. 1 shows a typical path of current measurement. Current signals, which are measured by current sensors, are converted into

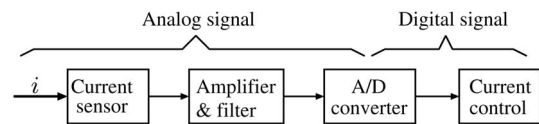


Fig. 1. Path of current measurement.

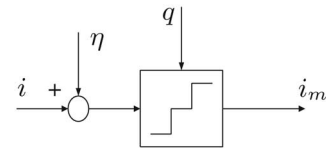


Fig. 2. Block diagram of current measurement.

digital signals via ADCs after the amplitude amplification and noise filtering. During the process, quantization error and metering noise containing scaling error, offset error, and thermal noise are introduced into the control system. The quantization error arises between the input and the output of ADCs because the analog signal may assume any value within the input range, while the output data are finite-precision samples [10]. This kind of error behaves as highly colored noise and cannot be simply suppressed by low-pass filter. In contrast, the scaling error and the offset error are usually caused by aging deterioration, and can be corrected by periodic calibration. In this paper, the current measurement system is assumed to be well calibrated and the scaling error and the offset error considered in [5] and [8] are ignored. Fig. 2 shows the block diagram of the measurement process, where η denotes the metering noise and q denotes the quantization error. An example to demonstrate the characteristic of current measurement error is shown in Fig. 3. Here, the original current signal i is a sinusoidal signal with the magnitude and frequency of 1 A and 5 Hz. The quantization step is 0.1 A and the metering noise is uniform in $[-0.025, 0.025]$ A. The measurement error and the corresponding power spectral density (PSD) are also shown in the figure. The harmonics caused by the quantization are observable all over the frequency domain. Torque ripples may therefore be generated by these harmonics. Usually, mechanical system can work as a low-pass filter to reduce the high-order harmonics. Low-order harmonics, however, cannot be effectively reduced. In order to suppress current quantization effects, high-resolution ADCs can be used to improve the quality of measured signals. However, not only would the implementation cost be increased, but also the sample rate of ADCs will be limited due to the trade-off between resolution and sample rate [10]. In [1],

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The authors are with the Department of Electrical Engineering and Information Systems, University of Tokyo, Kashiwa, Chiba 277-8561, Japan (e-mail: zhu@hflab.k.u-tokyo.ac.jp; fujimoto@k.u-tokyo.ac.jp).

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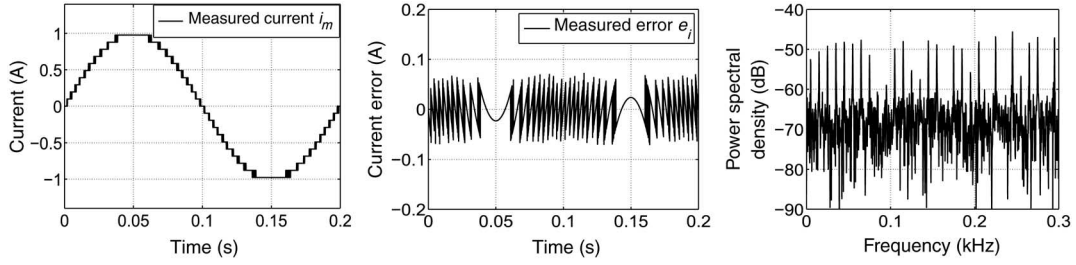


Fig. 3. Example of quantization effects. Left graph shows the measured current signal, middle graph shows the measurement error, and right graph shows the PSD of the measurement error.

Naouar *et al.* reported that A/D conversion time limits the current control performance when high-speed processor, such as field-programmable gate array (FPGA), is implemented.

Many numerical methods have been proposed for suppressing quantization effects. Model-based observers can estimate the original signals from the quantized measurements robustly if the quantization error is small [11], [12]. However, this condition may not be satisfied in many real situations. Additionally, some mathematical optimization methods have also been proposed to reconstruct the original signals [13], [14]. However, the computational time may be too long to be usable in many field applications, where the algorithm should be implemented in, e.g., digital signal processor (DSP).

Compared with quantization noise, white noise contains equal power at any center frequency and can be easily reduced. Many approaches, such as Linear-Quadratic-Gaussian (LQG) controller design method, are very effective and robust if the noise introduced into the system is white [15]. Motivated by this perspective, dithering techniques have been studied to whiten the quantization noise [16], [18], [19]. However, the additional noise may increase the wide-band noise level and result in no additional improvement.

Based on the above observation, a novel method combining dithering techniques and Kalman filter was proposed in [17]. The method can enhance the quality of the measured current signals significantly. However, the study only focuses on the case where the metering noise is Gaussian in the design of nonsubtractively dithered system. Other common cases, such as that the metering noise is uniform, are not analyzed. Moreover, the reasons why current quantization would excite the torque ripples are not clarified in this paper.

The purpose of this paper is to extend the research in [17]. The contributions of this paper are.

- 1) Not only the case when the metering noise is Gaussian, the case when the metering noise is uniform is also considered in the design of nonsubtractively dithered system.
- 2) Torque ripples generated by current quantization error is analyzed and evaluated.
- 3) A careful comparison of the proposed method with a commonly used method, namely, low-pass filter, is provided in simulations and experiments.

The remainder of this paper is organized as follows. Section II introduces the model of plant. The dithering techniques and Kalman filter for suppressing quantization effects are presented in Section III. Sections IV and V demonstrate the effectiveness of the proposed approach by simulations and experiments. Finally, the conclusion is given in Section VI.

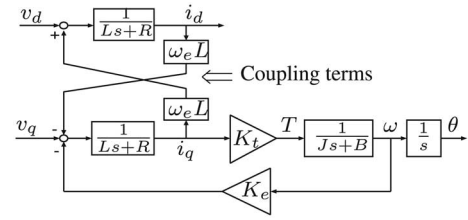


Fig. 4. Model of SPMSM.

II. SYSTEM DESCRIPTION

Current control for a surface permanent magnet synchronous motor (SPMSM) is considered. According to Park Transform, d, q -axis voltage equations are given in [5]

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} Ls + R & -\omega_e L \\ \omega_e L & Ls + R \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \omega K_e \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (1)$$

where s denotes the Laplace operator, $i_{d,q}$ denote the d, q -axis currents, $v_{d,q}$ denote the d, q -axis voltages, L is the stator inductance, R is the stator resistance, K_e is the back electromotive force (back-EMF) constant, and ω_e and ω are the field angular speed and the motor speed, respectively. The model is shown in Fig. 4. Defining v'_d and v'_q by

$$v'_d := v_d + \omega_e L i_q \quad (2)$$

$$v'_q := v_q - (\omega_e L i_d + \omega K_e), \quad (3)$$

equation (1) can be rewritten as

$$\begin{bmatrix} v'_d \\ v'_q \end{bmatrix} = \begin{bmatrix} Ls + R & 0 \\ 0 & Ls + R \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix}. \quad (4)$$

According to Park Transform, the following equation is obtained:

$$\begin{bmatrix} i_u \\ i_v \\ i_w \end{bmatrix} = \frac{2}{3} \frac{1}{Ls + R} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} v'_u \\ v'_v \\ v'_w \end{bmatrix} \quad (5)$$

where $[v'_u \ v'_v \ v'_w]^T$ is the fictive voltage vector calculated from $[v'_d \ v'_q]^T$.

In digital control, metering noise η and quantization error q are introduced into the control system, as shown in Fig. 2. An ideal ADC is a nonlinear device having a staircase-type I/O relation.

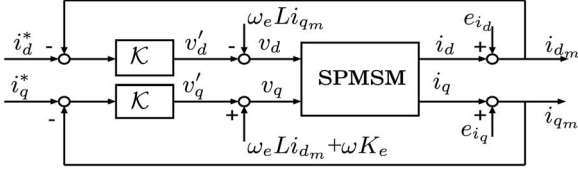


Fig. 5. Current control of SPMSM.

Assuming that the input is always within the measurement range of ADCs, the output y can be expressed in terms of the input x as

$$y = Q(x) = \Delta \left\lfloor \frac{x}{\Delta} + \frac{1}{2} \right\rfloor \quad (6)$$

where $Q(\cdot)$ is the quantization operator, $\lfloor \cdot \rfloor$ is the floor operator, and Δ is the quantization step defined by

$$\Delta = \frac{I_0}{2^{N_b-1}}. \quad (7)$$

Here, I_0 is the one-sided measurement range of current sensor and N_b is the resolution of the ADC. The quantization error is defined by

$$q(x) := Q(x) - x. \quad (8)$$

The quantization error q is dependent of x unless x satisfies the so-called band-limited condition [22]. In many cases, where Δ is small, $q(x)$ can be regarded as uniform noise in $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$ and the mean and variance are expressed as [18]

$$E[q] = 0 \quad (9)$$

$$E[q^2] = \frac{\Delta^2}{12} \quad (10)$$

where $E[\cdot]$ means the expectation.

Suppose that the proportional–integral (PI) current regulator with the pole-zero cancellation feature

$$\mathcal{K} = \frac{Ls + R}{\tau s} \quad (11)$$

is applied for current control. Here, τ determines the bandwidth of current feedback loop. Besides $d-q$ stator current components, decoupled control is also applied to obtain good dynamic and static control performances. Fig. 5 shows the current control block diagram. In the figure, $i_{d,qm}$ and $e_{i_d,q}$ are the d, q -axis currents calculated from the the measured currents and the d, q -axis currents measurement errors, respectively. Considering a three-phase SPMSM drive system with U,W-phase current sensors, $e_{i_{d,q}}$ can be expressed by [8]

$$e_{i_d} = \frac{2}{\sqrt{3}} \sqrt{e_{i_u}^2 + e_{i_u} e_{i_w} + e_{i_w}^2} \sin(\theta_e + \alpha) \quad (12)$$

$$e_{i_q} = \frac{2}{\sqrt{3}} \sqrt{e_{i_u}^2 + e_{i_u} e_{i_w} + e_{i_w}^2} \cos(\theta_e + \alpha) \quad (13)$$

where $e_{i_{u,w}}$ are the U,W-phase measurement errors including quantization errors $\alpha = \tan^{-1}(\sqrt{3}e_{i_u}/(e_{i_u} + 2e_{i_w}))$ and θ_e is the

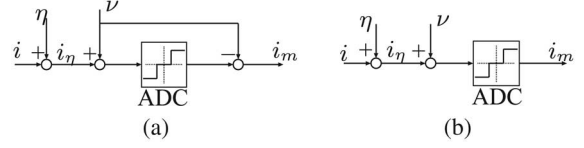


Fig. 6. Dithering method for current quantization. (a) Subtractively dithered method and (b) nonsubtractively dithered method.

field angular. According to the control system shown in Fig. 5, the torque T is expressed by

$$\begin{aligned} T &= K_t i_q = \frac{K_t}{\tau s + 1} \left(i_q^* - e_{i_q} + \omega_e L \frac{\tau s}{Ls + R} e_{i_d} \right) \\ &= \frac{K_t}{\tau s + 1} i_q^* + \Delta T \end{aligned} \quad (14)$$

where

$$\Delta T = \frac{K_t}{\tau s + 1} \left(-e_{i_q} + \omega_e L \frac{\tau s}{Ls + R} e_{i_d} \right). \quad (15)$$

Although a motor and current control loop can work as low-pass filters to reduce high-order harmonics inherent in $e_{i_{u,w}}$, low-order harmonics may also induce torque ripple according to (12), (13), and (15).

In Section III, dither techniques are proposed as the first step to whiten the quantization error. Then, Kalman filter is designed to estimate the real current signals from the dithered measurements.

III. DITHERING TECHNIQUES AND KALMAN FILTER FOR CURRENT QUANTIZATION

Dither is an intentionally applied form of noise used to decorrelate signal-dependent noise. There are two archetypes of dithered systems: 1) subtractively dithered system, where the dither is subsequently subtracted from the output signal after quantization; and 2) nonsubtractively dithered system, where the dither is not subtracted from the output signal. In the following, subtractively dithered system and nonsubtractively dithered system are firstly designed for whitening the current quantization noise in Sections III-A and III-B, respectively. Then, Kalman filter for estimating the real current signals from the dithered current signals is given in Section III-C.

A. Subtractively Dithered System

Subtractively dithered system for suppressing current quantization error is shown in Fig. 6(a). The signal ν is the dither to be designed. The measurement error can be expressed by

$$\begin{aligned} e_i &= i_m - i \\ &= Q(i_\eta + \nu) - (i_\eta + \nu) + \eta \\ &= q(i_\eta + \nu) + \eta. \end{aligned} \quad (16)$$

According to *Schuchman's Condition* [19], $q(i_\eta + \nu)$ is independent of i_η if ν is uniform in $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$. Suppose η is white noise, the mean and variance of e_i are calculated by [17]

$$E[e_i] = 0 \quad (17)$$

$$E[e_i^2] = E[\eta^2] + \frac{\Delta^2}{12}. \quad (18)$$

The subtractively dithered system is ideal in the sense that not only can e_i be rendered independent of the input, the noise level is also not enlarged even after introducing the dither ν [the variance (18) is the same as (10) when $E[\eta^2]$ is ignored].

B. Nonsubtractively Dithered System

Subtractively dithered system requires synchronous dither subtraction at the system output, which may not always be implementable in practical situations. For such reason, the non-subtractively dither technique in the following is also of interest.

Nonsubtractively dithered system for current quantization is shown in Fig. 6(b). The current measurement error is expressed by

$$\begin{aligned} e_i &= i_m - i \\ &= Q(i_\eta + \nu) - i_\eta + \eta \\ &= q(i_\eta + \nu) + \nu + \eta. \end{aligned} \quad (19)$$

According to the *Theorem* in [20], a dither with the triangular probability density function (PDF)

$$p_\nu(\epsilon) = \begin{cases} \frac{1}{\Delta^2}(\epsilon + \Delta), & -\Delta < \epsilon \leq 0 \\ \frac{1}{\Delta^2}(\epsilon - +\Delta), & 0 < \epsilon \leq \Delta \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

is the unique choice to render the mean and the variance of $q(i_\eta + \nu) + \nu$ independent of the input i_η with a minimal noise level. In this case, mean and variance of ν and e_i are expressed as:

$$E[\nu] = 0 \quad (21)$$

$$E[\nu^2] = \frac{\Delta^2}{6} \quad (22)$$

$$E[e_i] = 0 \quad (23)$$

$$E[e_i^2] = E[\eta^2] + \frac{\Delta^2}{4}. \quad (24)$$

Compared to (18), $E[e_i^2]$ is degraded and may be too large to be available. The probability distribution characteristics of η is utilized to reduce $E[e_i^2]$ as follows.

For convenience, w is defined as the sum of dither ν and metering noise η as follows:

$$w := \eta + \nu. \quad (25)$$

According to (19), the measurement error e_i is expressed by

$$e_i = q(i + w) + w. \quad (26)$$

In order to render e_i independent of i and minimize the total measurement noise, w is required to be the triangular noise expressed by (20). According to the Convolution Theorem and Inverse Fourier Transform, a theoretical selection of ν is that p_ν should satisfy

$$p_\nu = \mathcal{F}^{-1} \frac{\text{sinc}^2(\pi\Delta u)}{P_\eta(u)} du. \quad (27)$$

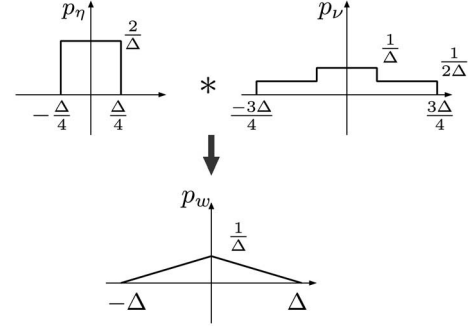


Fig. 7. Example of *Proposition 1* in the case of $N = 2$. Operator “*” means convolution.

Here, $\text{sinc}^2(\pi\Delta u)$ and $P_\eta(u)$ are the characteristic functions of the triangular PDF noise (20) and η , respectively. If ν satisfies (27), the variance of e_i is calculated by

$$E[e_i^2] = \frac{\Delta^2}{4}. \quad (28)$$

This implies that the variance of e_i is not affected by η and smaller than (24).

Usually, the dither ν with the PDF calculated by (27) is very hard to be generated in practical situations. Two common cases that η is uniform noise and η is Gaussian noise are examined as follows.

1) *The Metering Noise is Uniform Noise:* In this case, the following result is obtained.

Proposition 1: Suppose that η is uniform noise with the PDF

$$p_\eta(\epsilon) = \begin{cases} \frac{N}{\Delta}, & -\frac{\Delta}{2N} < \epsilon \leq \frac{\Delta}{2N} \\ 0, & \text{otherwise} \end{cases} \quad (29)$$

where $N \in \mathcal{N}$, then the PDF of w is the triangular PDF (20) if the PDF of ν satisfies

$$p_\nu(\epsilon) = \begin{cases} \frac{1}{N\Delta}, & \begin{cases} -\frac{2N-1}{2N}\Delta < \epsilon \leq -\frac{2N-3}{2N}\Delta \\ \frac{2N-3}{2N}\Delta < \epsilon \leq \frac{2N-1}{2N}\Delta \end{cases} \\ \frac{2}{N\Delta}, & \begin{cases} -\frac{2N-3}{2N}\Delta < \epsilon \leq -\frac{2N-5}{2N}\Delta \\ \frac{2N-5}{2N}\Delta < \epsilon \leq \frac{2N-3}{2N}\Delta \end{cases} \\ \vdots & \vdots \\ \frac{i}{N\Delta}, & \begin{cases} -\frac{2(N-i)+1}{2N}\Delta < \epsilon \leq -\frac{2(N-i)-1}{2N}\Delta \\ \frac{2(N-i)-1}{2N}\Delta < \epsilon \leq \frac{2(N-i)+1}{2N}\Delta \end{cases} \\ \vdots & \vdots \\ \frac{N-1}{N\Delta}, & \begin{cases} -\frac{3}{2N}\Delta < \epsilon \leq -\frac{1}{2N}\Delta \\ \frac{1}{2N}\Delta < \epsilon \leq \frac{3}{2N}\Delta \end{cases} \\ \frac{1}{\Delta}, & -\frac{1}{2N}\Delta < \epsilon \leq \frac{1}{2N}\Delta \\ 0, & \text{otherwise} \end{cases} \quad (30)$$

where $i = 1, 2, \dots, N-1$.

Proposition 1 can be proved according to the Convolution Theorem. Fig. 7 illustrates the convolutions of p_η and p_ν in the case of $N = 2$.

In the case where the condition (29) is not satisfied, say η is uniform in $[-\frac{\Delta}{2(K+\alpha)}, \frac{\Delta}{2(K+\alpha)}]$ for some $K \in \mathcal{N}$ and $-0.5 < \alpha \leq 0.5$, the principle of proximity can be applied for p_η .

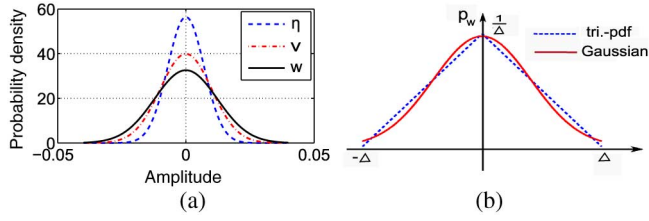


Fig. 8. (a) Illustration of $w = \eta + \nu$ from the view of probability distribution when η and ν are Gaussian and (b) comparison of triangular PDF and Gaussian distribution.

TABLE I
PARAMETERS OF SPMSM FOR SIMULATION

Rated current A	25.0	A
Rated Torque T	17.9	N·m
Inductance L	7.2	mH
Resistance R	2.16	Ω
Inertia J	0.001	kg·m ²
Viscosity B	0.01	N·m/(rad/s)
Thrust coefficient K_t	0.715	N·m/A
Back-EMF constant K_e	74.9	mV/(rad/s)

2) *Metering Error is Gaussian Noise*: In this case, the optimal design of ν by (27) is unavailable. Therefore, a suboptimal design of ν is considered. The following proposition is achieved:

Proposition 2: Suppose that the metering error η is Gaussian noise and has the Normal distribution

$$p_\eta(\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}\epsilon^2\right\}, \quad 0 < \sigma^2 < \frac{\Delta^2}{6} \quad (31)$$

where σ is the standard deviation. Then, w is Gaussian with

$$p_w(\epsilon) = \frac{1}{\sqrt{2\pi\frac{\Delta^2}{6}}} \exp\left\{-\frac{1}{2\frac{\Delta^2}{6}}\epsilon^2\right\} \quad (32)$$

and has the same variance with the triangular dither (20) if p_ν satisfies

$$p_\nu(\epsilon) = \frac{1}{\sqrt{2\pi(\frac{\Delta^2}{6} - \sigma^2)}} \exp\left\{-\frac{1}{2\frac{\Delta^2}{6} - \sigma^2}\epsilon^2\right\}. \quad (33)$$

Proposition 2 can be proved according to the Convolution Theorem. In the case of $\sigma^2 \geq \frac{\Delta^2}{6}$, η can be regarded to be large enough to whiten the quantization error, so that ν becomes unnecessary. Fig. 8(a) illustrates the sum of two variables from the view of probability perspective. By calculation, the cumulative probability of (32) located in $[-\Delta, \Delta]$ is about 98.6%.

The comparison between the Normal distribution and the triangular distribution with the same variance is shown in Fig. 8(b). For practical situation, a dither with Normal distribution has almost the same advantage to whiten the quantization error. A further insight into the discussion is given in [21].

Even the metering noise can be exploited to reduce the total noise level in nonsubtractively dithered system, the additional

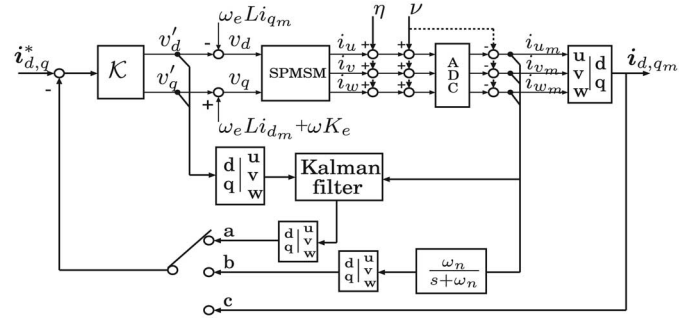


Fig. 9. Block diagram of current control for simulations.

noise may result in no improvement in current control. In Section III-C, Kalman filter is designed to estimate the original current signals from the dithered signals.

C. Kalman Filter for Noise Suppression

A state-space realization of the model (5) is expressed as [17]

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (34)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} \quad (35)$$

where $\mathbf{x} = [i_u \ i_v \ i_w]^T$, $\mathbf{u} = [v_u' \ v_v' \ v_w']^T$ and

$$\mathbf{A} = \text{diag}\left(-\frac{R}{L}, -\frac{R}{L}, -\frac{R}{L}\right)$$

$$\mathbf{B} = \frac{2}{3L} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix}$$

$$\mathbf{C} = \mathbf{I}_3$$

where $\mathbf{I}_n$ denotes the identity matrix with size n . Based on this model, Kalman filter is formulated by

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}_k(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}}) \quad (36)$$

where $\hat{\mathbf{x}}$ is the estimation of $\mathbf{x}$ and $\mathbf{L}_k$ is the Kalman gain calculated via

$$\mathbf{L}_k = \mathbf{P}\mathbf{C}^T\mathbf{R}_c^{-1}. \quad (37)$$

Here, $\mathbf{R}_c$ is the covariance of measurement noise and $\mathbf{P}$ is the solution of Riccati equation

$$\mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^T - \mathbf{P}\mathbf{C}^T\mathbf{R}_c^{-1}\mathbf{C}\mathbf{P} + \mathbf{Q}_c = \mathbf{0} \quad (38)$$

where $\mathbf{Q}_c$ is the covariance of the input disturbance. When dithering techniques proposed in Sections III-A and III-B are applied, the variance of the e_i can be calculated accurately, and, therefore, $\mathbf{R}_c$ can be analytically determined.

IV. SIMULATIONS

In this section, simulations are performed to verify the effectiveness of the proposed dithering techniques and Kalman filter. The parameters of an SPMSM used for simulations are shown in Table I. For comparison, the case that measured d, q -axis currents

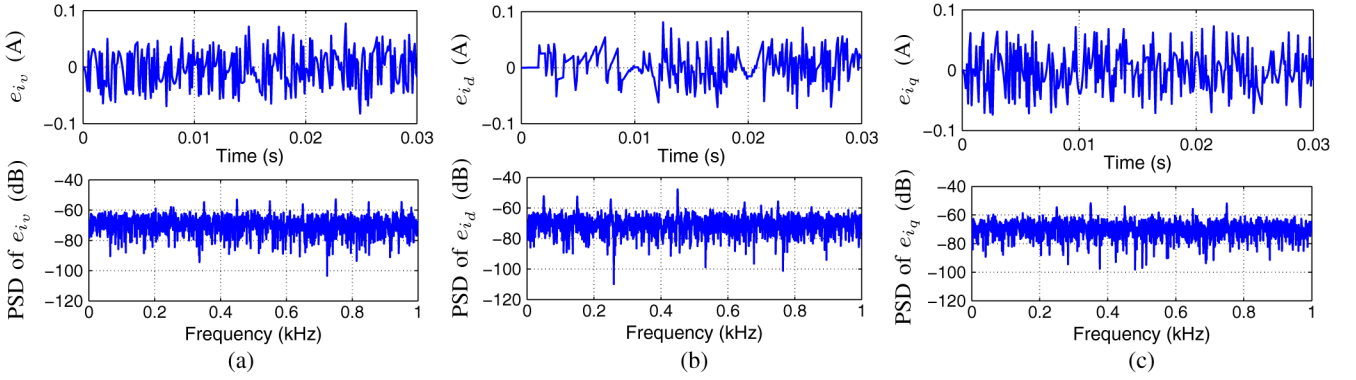


Fig. 10. Simulation results of Case 1 without dither and without Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

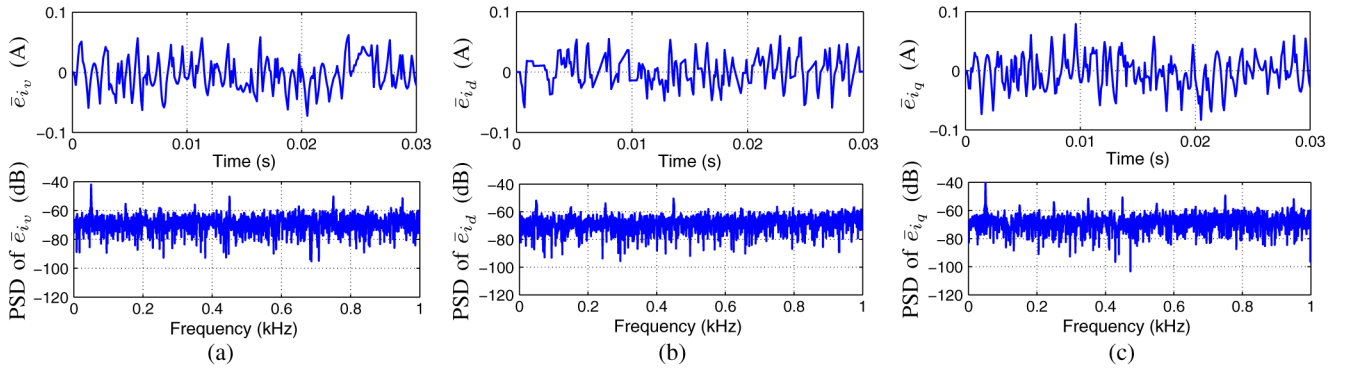


Fig. 11. Simulation results of Case 2 without dither and with low-pass filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

are used directly as feedback and the case that current signals filtered by a low-pass filter are used as feedback are implemented. The control block diagram is shown in Fig. 9. The bandwidth of current control loop is set as 1 kHz. MATLAB/Simulink is used for performing the simulations.

For evaluation, the following errors are defined:

$$e_{i_v} := i_{v_m} - i_v \quad (39)$$

$$\hat{e}_{i_v} := \hat{i}_v - i_v \quad (40)$$

$$\hat{e}_{i_d} := \hat{i}_d - i_d \quad (41)$$

$$\hat{e}_{i_q} := \hat{i}_q - i_q \quad (42)$$

$$\bar{e}_{i_v} := \bar{i}_v - i_v \quad (43)$$

$$\bar{e}_{i_d} := \bar{i}_d - i_d \quad (44)$$

$$\bar{e}_{i_q} := \bar{i}_q - i_q. \quad (45)$$

Here, e_{i_v} is the measured V-phase current error, $\hat{i}_{v,d,q}$ are the estimated current signals by Kalman filter, and $\bar{i}_{v,d,q}$ are the filtered current signals by the low-pass filter.

In the setup, ADCs with a resolution of 10-bit and current sensors with a measurement range of ± 50 A are used. The resulting quantization step is $\Delta = 0.0977$ A. The metering noise η is set as uniform noise in $[-\frac{\Delta}{4}, \frac{\Delta}{4}]$. The covariance matrix $\mathbf{R}_c$ for Kalman filter is set as $\mathbf{R}_c = \text{diag}(E[e_{i_v}^2], E[e_{i_d}^2], E[e_{i_q}^2])$. Here, $E[e_{i_v}^2]$, $E[e_{i_d}^2]$, $E[e_{i_q}^2]$ are determined by (18) if the subtractively dithering method is applied, or by (28) if the nonsubtractively dithering method is applied. The covariance matrix $\mathbf{Q}_c$

is properly set as $\mathbf{Q}_c = \text{diag}(0.01, 0.01, 0.01)$. The bandwidth of the first-order low-pass filter shown in Fig. 9 is set as 1.5 kHz.

In this section, the following six cases are examined:

- 1) Case 1: without dither and without Kalman filter;
- 2) Case 2: without dither and with low-pass filter;
- 3) Case 3: with subtractive dither and without Kalman filter;
- 4) Case 4: with nonsubtractive dither and without Kalman filter;
- 5) Case 5: with subtractive dither and with Kalman filter;
- 6) Case 6: with nonsubtractive dither with Kalman filter.

The reference signals are set as $i_q^* = 0.5 \sin 314t$ and $i_d^* = 0$, and the simulation results are shown in Figs. 10–16. PSD of current errors and torque errors is exploited to evaluate the performance of the cases. In Case 1, the harmonics caused by quantization exist not only in phase current signals but also in d , q -axis current signals. The high-order harmonics can be somewhat suppressed by low-pass filter, as shown in Fig. 11. However, the noise in low-frequency domain is not reduced. Additionally, the phase delay caused by the filter also excites the first-order resonance. On the other hand, the harmonics are reduced when dithering techniques are applied, as shown in Figs. 12 and 13. These results also indicate that current quantization errors are whitened by both the subtractively dithered method and the nonsubtractively dithered method. Although the average noise level is augmented by nonsubtractively dithered method, the maximum value of PSD is reduced drastically.

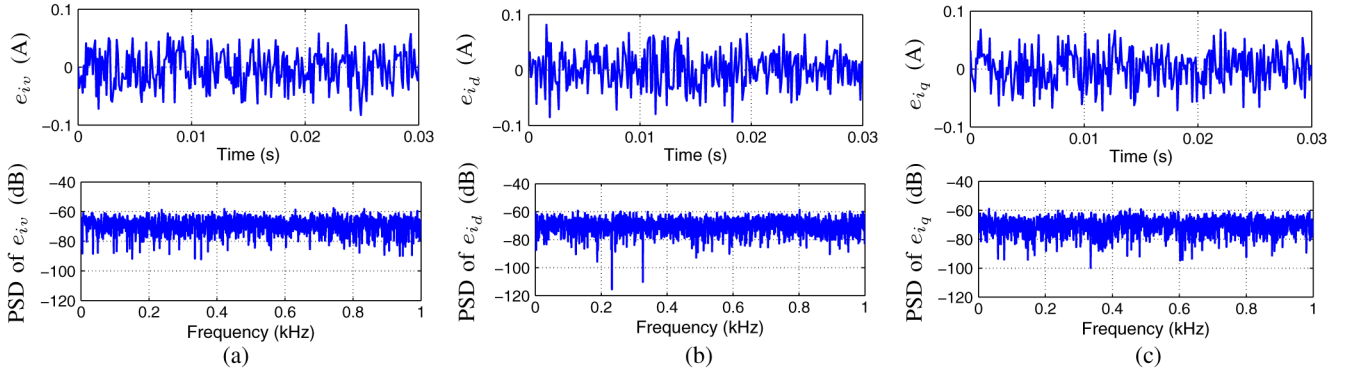


Fig. 12. Simulation results of Case 3 with subtractive dither and without Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

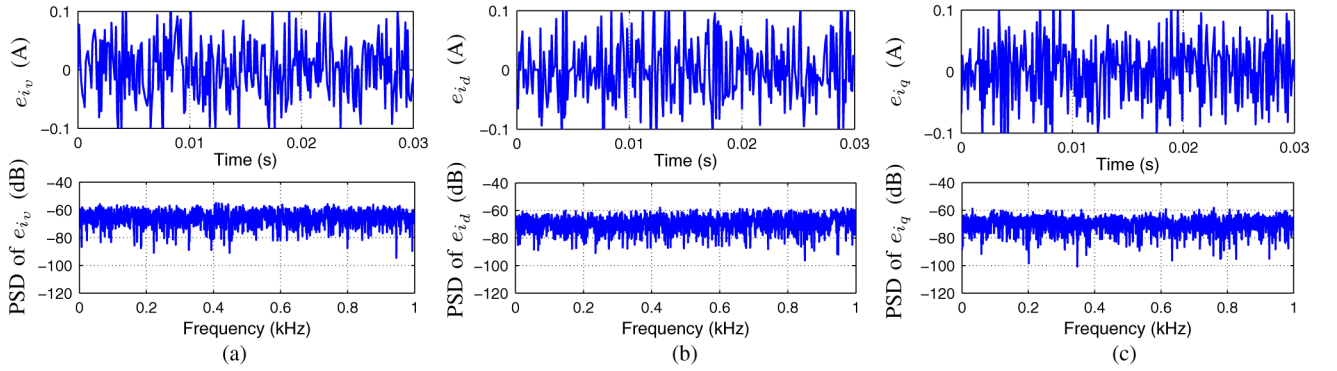


Fig. 13. Simulation results of Case 4 with nonsubtractive dither and without Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

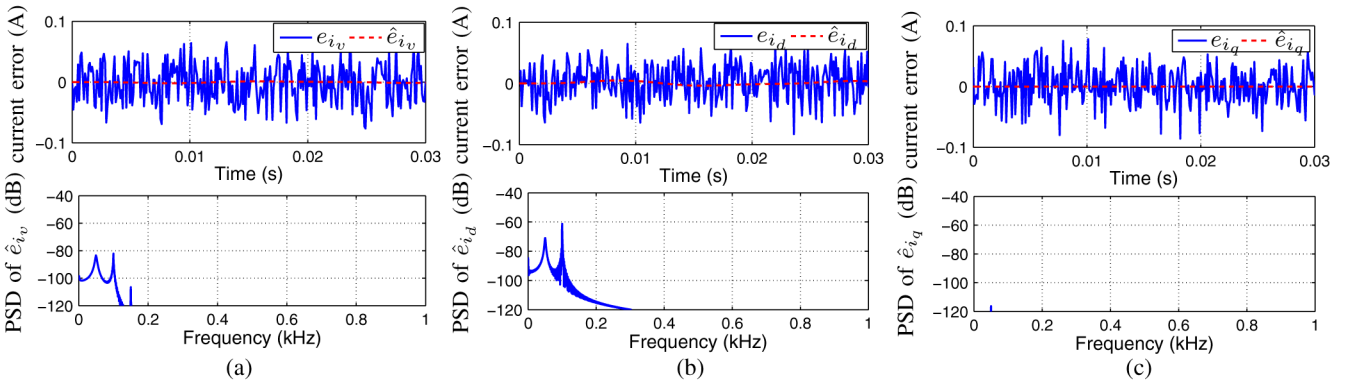


Fig. 14. Simulation results of Case 5 with subtractive dither as well as Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

The results of dithering techniques combined with Kalman filter are shown in Figs. 14 and 15. The estimation errors $\hat{e}_{i_{v,d,q}}$ are far smaller than the measurement errors $e_{i_{v,d,q}}$, which means that the designed Kalman filter can estimate the current signals with high accuracy. Since Kalman filter also introduces phase delay, first-order resonances are also introduced. However, compared to the results of Case 2, the peak values of the resonances are relatively small. The comparison of the current errors evaluated

via root mean square (RMS) and maximum value of PSD is shown in Table II.

The characteristics of the torque error of each case calculated by (15) is shown in Fig. 16. The torque ripples cannot be suppressed by low-pass filter. Moreover, the phase delay introduced by the filter worsens the torque ripples, as shown in Fig. 16(b). Fig. 16(c) and (d) show that the torque error can also be whitened by dithering techniques. In addition, the torque error

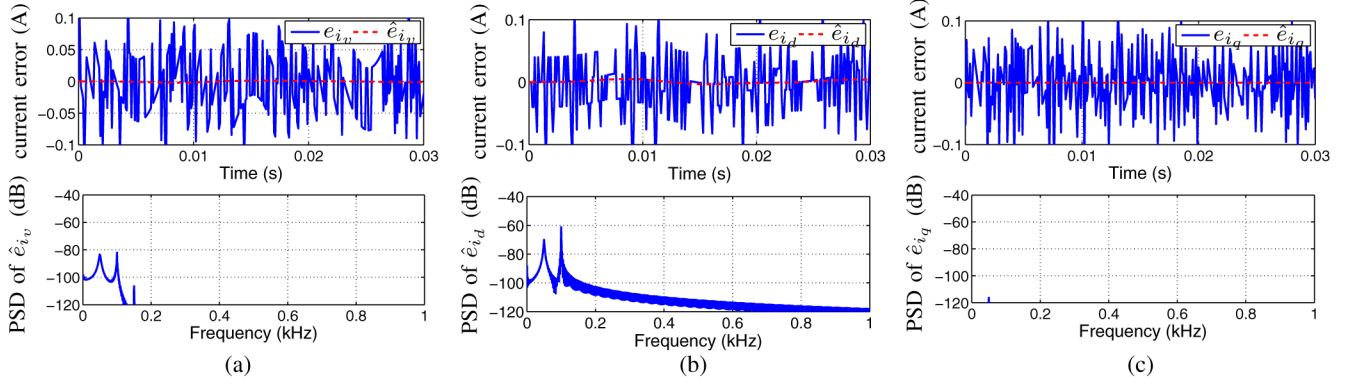


Fig. 15. Simulation results of Case 6 with nonsubtractive dither as well as Kalman filter. (a) Comparison of V-phase current. (b) Comparison of d -axis current. (c) Comparison of q -axis current.

TABLE II
COMPARISON OF THE CURRENT MEASUREMENT ERRORS (SIMULATION RESULTS)

Situations		Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
$e_{i_v}/\bar{e}_{i_v}/\hat{e}_{i_v}$	RMS ($\times 10^{-2}$)	3.05	2.38	3.14	4.91	$2.20e^{-2}$	$2.42e^{-2}$
	PSD _{peak} [dB]	-52.9	-41.4	-57.7	-55.0	-82.2	-81.6
$e_{i_d}/\bar{e}_{i_d}/\hat{e}_{i_d}$	RMS ($\times 10^{-2}$)	3.33	2.69	3.17	4.91	0.113	0.120
	PSD _{peak} [dB]	-47.8	-51.9	-58.7	-58.2	-61.2	-61.0
$e_{i_q}/\bar{e}_{i_q}/\hat{e}_{i_q}$	RMS ($\times 10^{-2}$)	3.11	2.45	3.14	4.93	$2.45e^{-3}$	$2.96e^{-3}$
	PSD _{peak} [dB]	-53.5	-42.6	-58.9	-57.9	-116.3	-116.1

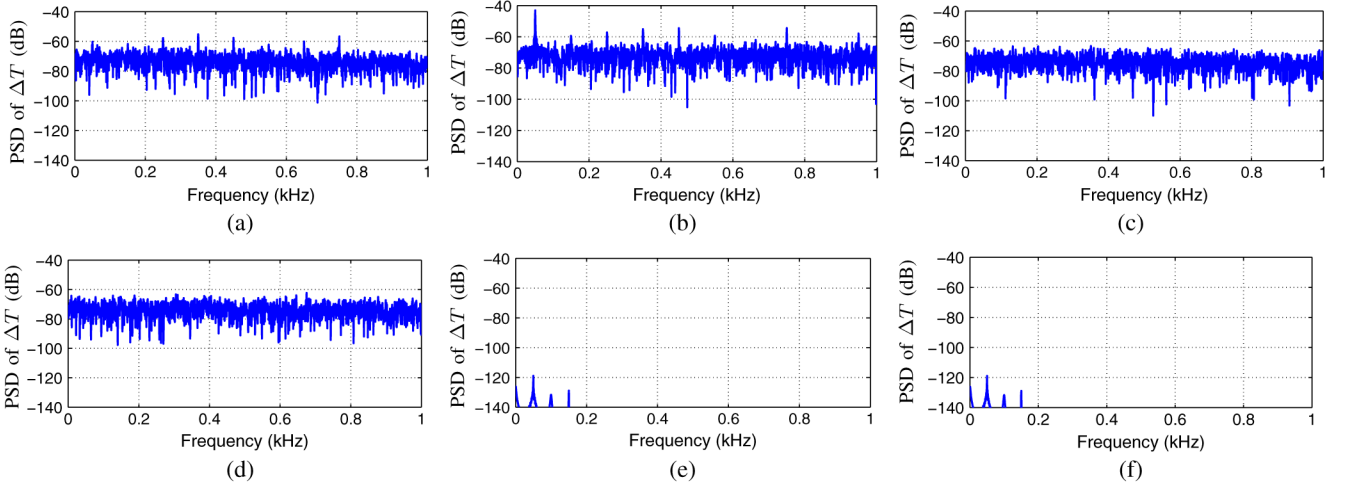


Fig. 16. Torque ripples generated by current measurement error: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4; (e) Case 5; and (f) Case 6.

TABLE III
COMPARISON OF THE TORQUE ERRORS (SIMULATION RESULTS)

Situations	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
$\Delta T_{\text{PSD peak}}$ [dB]	-55.27	-43.05	-63.30	-62.27	-118.9	-118.2

is significantly reduced by the methods combining dithering techniques and Kalman filter, as shown in Fig. 16(e) and (f). The comparison of the torque errors evaluated via the maximum value of PSD is shown in Table III.

From the results mentioned above, the combination of subtractively dithered technique and Kalman filter obtains the best performance. In the case where the synchronous dither subtraction is unavailable, the combination of nonsubtractively dithered technique and Kalman filter is desirable.

V. EXPERIMENTS

A. Description of the Experiment System

A high-precision stage used for experiments is shown in Fig. 17, and its parameters are shown in Table IV. The stage is a simplified model of the scanners in exposure systems used for the fabrication of integrated circuits. Two linear motors located at the both sides of the carriage are applied to drive the stage. An air guide system is introduced to reduce the friction between the

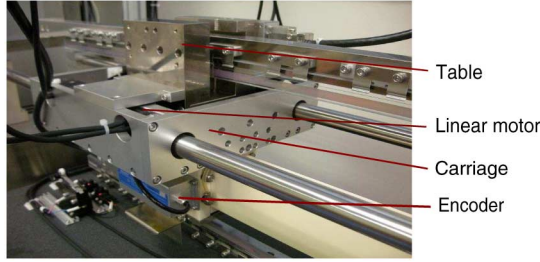


Fig. 17. Experimental device.

TABLE IV
PARAMETERS OF STAGE

Rated current $I_{\max}$	0.54	A
Rated thrust $F_{\max}$	14.3	N
Inductance L	15.5	mH
Resistance R	20.5	Ω
Mass M	14.7	kg
Viscosity B	24	N/(m/s)
Thrust coefficient K_t	26.5	N/A
Back-EMF constant K_e	16	V/(m/s)

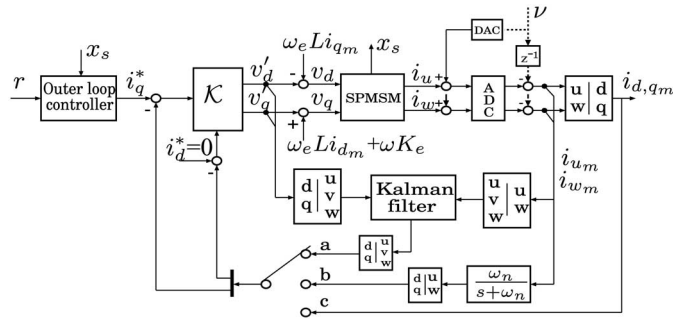


Fig. 18. Block diagram of control system.

stator and the slider of the motors. U, W-phase current signals are measured by FA-050PV (LEM) with a measurement range of ± 50 A. The signals are then converted into digital signals via high-precision ADCs with a resolution of 14-bit. DSP (TMS320C6713, 225 MHz) is used as the processor to implement the proposed approaches.

The bandwidth of current control loop is set as 1 kHz. A two-degree-of-freedom controller is exploited for position control [23], and the resulting bandwidth of position control loop is 10 Hz. The block diagram of control system is shown in Fig. 18.

In order to verify the effectiveness of the dithering techniques, the resolution of the ADCs is dropped to 10-bit by software, so that the resulting current quantization step is $\Delta = 0.0977$ A. For comparison, a precise current probe (Tektronix) is used to measure the real current signal. Dither signals are generated in DSP by “Box–Muller method” [24] and then converted into analog signals via D/A converters with a resolution of 16-bit.

First, the probability distribution of the metering noise is investigated. A current signal measured by current measurement system without executing the control system is used for analyzing the probability characteristic of the metering noise η . A sample of η in time domain is shown in Fig. 19(a) and the histogram plot is shown in Fig. 19(b). The metering noise can be approximated with

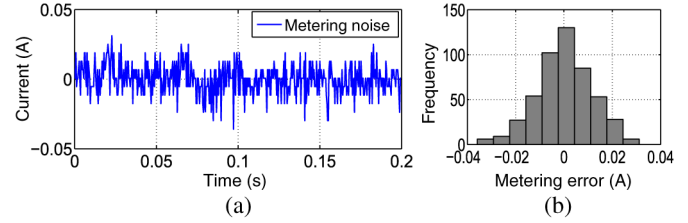


Fig. 19. (a) Example of the metering noise and (b) its histogram analysis.

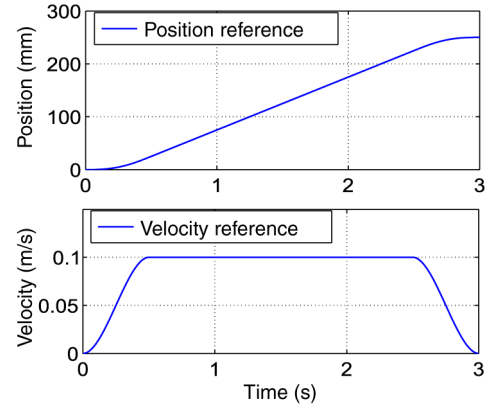


Fig. 20. Position and velocity reference.

Gaussian noise. By calculation, the variance of η is about $E[\eta^2] = 1.1218 \times 10^{-4}$. Based on this value, the nonsubtractive dither is designed according to (33). The covariance matrix R_c for Kalman filter is designed based on the error variances (18) and (28). Q_c is properly set as $Q_c = \text{diag}(0.05, 0.05, 0.05)$.

B. Experimental Evaluation

The trajectory reference including acceleration, uniform motion, and deceleration is given, as shown in Fig. 20. Six cases defined in Section IV are performed for evaluation. For comparison, the position tracking error for evaluation is defined as follows:

$$e_s := r - x_s \quad (46)$$

where r is the reference trajectory and x_s is the position of the stage.

Fig. 21 shows the results of Cases 1–4. For simplicity, only U-phase currents are used for analysis. In Case 1, the harmonics caused by quantization appear especially at low frequencies, as shown in Fig. 21(a). From Fig. 21(b), although the level of measurement noise can be somewhat reduced by low-pass filter, the harmonics cannot be suppressed. In contrast, the PSD of e_{iu} is flat all over the frequency domain when dithering techniques are applied, as shown in Fig. 21(c) and (d). These results indicate that the quantization error is whitened by the dithering techniques. Fig. 22 shows the results of Cases 5 and 6. The estimation errors are much smaller than the measurement errors. These results indicate that the estimated currents are very close to real current signals. The comparison of position tracking performance in Cases 1, 3, and 4 without filter is shown in Fig. 23, and the comparison of control performance of Cases 2, 5, and 6

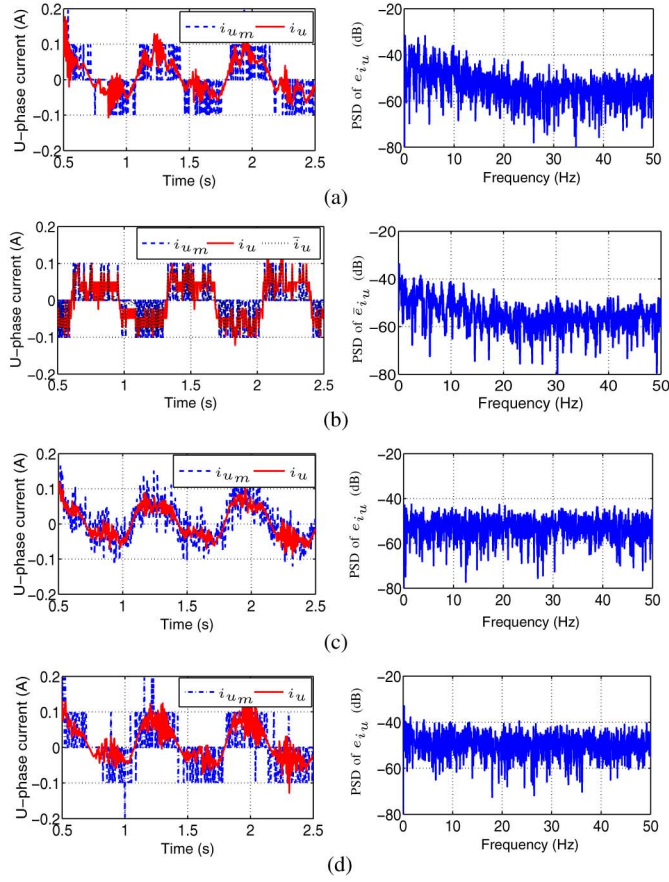


Fig. 21. Experimental results of Case 1-4. Graph (a) shows the results of Case 1 that the harmonics caused by quantization appear especially at low frequencies. Graph (b) shows the results of Case 2 that although the level of measurement noise can be somewhat reduced by low-pass filter, the harmonics cannot be suppressed. Graphs (c) and (d) show the results of Case 3 and Case 4, respectively. The results indicate that the harmonics caused by quantization noise are suppressed by the dithering techniques.

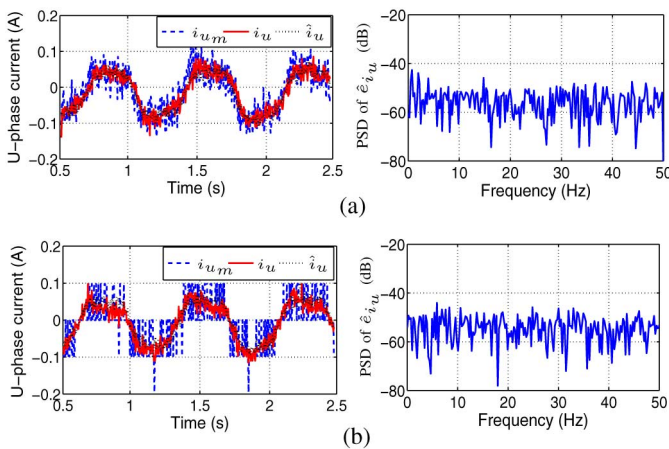


Fig. 22. Experimental results without applying Kalman filter. (a) Experimental results of Case 5. (b) Experimental results of Case 6.

with low-pass filter, dither techniques, and Kalman filter, respectively, is shown in Fig. 24. The maximum position tracking errors of all six cases are summarized in Table V. From the results, the combination of subtractively dithered technique and Kalman filter can obtain minimum position tracking error.

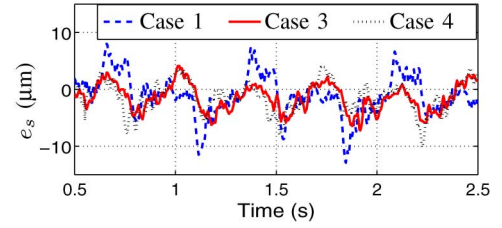


Fig. 23. Comparison of position tracking errors of Cases 1, 3, and 4, where no filter is applied.

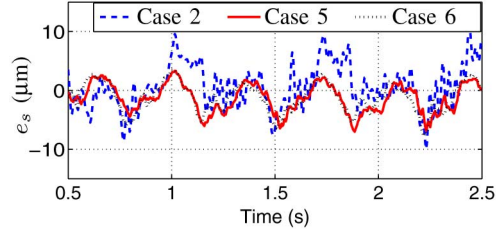


Fig. 24. Comparison of position tracking errors of Cases 2, 5, and 6, where filters are applied.

TABLE V
COMPARISON OF POSITION TRACKING ERROR (EXPERIMENTAL RESULTS)

Case	1	2	3	4	5	6
RMS of e_s [μm]	3.81	3.78	2.83	3.01	2.78	2.79
Max. of e_s [μm]	12.90	10.74	7.23	10.21	7.03	7.81

VI. CONCLUSION

In this paper, the combination of dithering techniques and Kalman filter for suppressing current quantization effects is proposed. In order to whiten the quantization error and reduce the total measurement noise level simultaneously, the probability characteristics of the metering noise is utilized to design the nonsubtractively dithered system. Furthermore, Kalman filter is applied to estimate the real current signals from the dithered measurement signals. Since the variance of the total measurement error is calculable, the covariance matrix for Kalman filter becomes explicit. The simulation and experimental results demonstrate the effectiveness of the proposed method.

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Hongzhong Zhu (S'11) received the B.S. degree in mechanical engineering from Shanghai University of Electric Power, Shanghai, China, in 2003, and the M.S. degree in mechanical systems control from the Department of System Science, Kyoto University, Kyoto, Japan, in 2011. He is currently working toward the Ph.D. degree in electrical engineering with the Department of Electrical Engineering and Information Systems, Graduate School of Engineering, University of Tokyo, Tokyo, Japan.

From 2003 to 2008, he was an Engineer with the Beilun Power Plant, China Guodian Corporation, Ningbo, China. His research interests include high-precision motion control, robust control, and convex optimization.

Mr. Zhu is a Member of the Institute of Electrical Engineerings of Japan.



Hiroshi Fujimoto (S'99–M'01–SM'12) received the Ph.D. degree in electrical engineering from the Department of Electrical Engineering, University of Tokyo, Tokyo, Japan, in 2001.

In 2001, he joined the Department of Electrical Engineering, Nagaoka University of Technology, Niigata, Japan, as a Research Associate. From 2002 to 2003, he was a Visiting Scholar at the School of Mechanical Engineering, Purdue University, West Lafayette, IN, USA. In 2004, he joined the Department of Electrical and Computer Engineering, Yokohama National University, Yokohama, Japan, as a Lecturer and went on to become an Associate Professor, in 2005. He is currently an Associate Professor at the University of Tokyo, since 2010. His research interests include control engineering, motion control, nano-scale servo systems, electric vehicle control, and motor drives.

Dr. Fujimoto received the Best Paper Award from IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS in 2001, the Isao Takahashi Power Electronics Award in 2010, and the Best Author Prize of the Society of Instrument and Control Engineers (SICE) in 2010. He is a Member of the Society of Instrument and Control Engineers, the Robotics Society of Japan, and the Society of Automotive Engineers of Japan.